SEION MATH CORE V5: SOFTWARE, EVIDENCE, AND REPRODUCIBILITY FOR PROJECTED MULTILINEAR GRAPHS

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Abstract. This companion documents the repository architecture, evidence contracts, test gates, paper-generation workflow, and governance state for the finite projected multilinear tree and graph program. The package separates executable behavior, mathematical claims, numerical artifacts, experiments, durable memory, and publication outputs. The v5 research layer adds an exact independent-law $k = 2$ sharpness construction, $k = 3$ lower witnesses, source-polynomial DAG propagation, signed-expression certificates, validated finite norm enclosures, and approximate-law budgets. The latest focused suite contains 85 passing tests; governance passes structurally with known historical warnings. The package remains fail-closed for theorem novelty, independent human review, and global sharpness.

1. Repository and evidence separation

The repository follows a source-of-truth policy. Executable behavior lives in `src/` and tests. Mathematical status lives in theorem and claim registries, proof dossiers, and counterexample records. Executed numerical facts live in manifests, metrics, certificates, and artifact hashes. Experiments live in registered matrices. Durable project state lives in `.ai/`. Generated paper tables and figures are derived outputs with provenance, never manual authorities.

This separation prevents a passing software test, a polished PDF, or a generated metric from silently becoming a universal mathematical statement. The governance contract distinguishes declared, observed, verified, and approved authority levels; human approval remains a separate gate.

2. Finite-core lineage

The finite v4 mathematical/software core is frozen at scientific commit `1f4984ec`; the complete SHA is preserved in the package manifest. The v5 theorem track then records the following immutable checkpoints:

checkpoint	role	scope
`10078f4`	V5 k=2 construction	independent-law saturation and equality audit
`773fa9c`	V5 freeze metadata	scientific baseline and ledger binding
`5e8384a`	V5-A construction	independent-law k=3 chain/branch lower witnesses
`997e745`	V5-A registry	lower-bound and conjecture status registration

The operational branch is `campaign/gate13-closeout`. Gate 13.5, Gate 14, KGR, and historical run trees are preserved and are not part of the v5 paper-generation change.

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Key words and phrases. research software, reproducibility, artifact contract, projected multilinear graph, governance.

Software and reproducibility companion. This paper does not constitute theorem approval, novelty approval, or release authorization.

3. Source architecture

The research implementation is intentionally split by generation:

path	responsibility
`src/seion_core/research_v3`	typed trees, exact/projected evaluation, nodewise certificates, enumeration
`src/seion_core/research_v4`	scalar DAG, source-aware, higher-order, signed, norm, and approximate-law layers
`src/seion_core/research_v5`	theorem-level k=2 sharpness, equality audit, and k=3 independent-law witnesses
`research/projected_trees_v4`	finite-core proof dossiers and registries
`research/projected_trees_v5`	V5 truth ledger, manuscript consolidation, formal targets, and novelty matrix
`claims/`	theorem, claim, counterexample, and novelty authority registries
`.ai/`	durable memory, tasks, decisions, blockers, evidence ledger, and handoff history

4. Verification gates

The focused v5 validation command is:

```
python -m pytest tests/research_v3 tests/research_v4 \\
  tests/research_v5_test_k2_sharpness.py \\
  tests/research_v5_test_equality_conditions.py \\
  tests/research_v5_test_k3_independent_candidates.py -q
```

It currently reports 85 passing tests. These include exact subset expansion, projected-root geometry, scalar and source-aware DAG certificates, repeated source multi-indices, signed cancellation, norm metadata, approximate-law budgets, k=2 exact saturation, equality compatibility, and k=3 lower-witness checks.

5. Artifact and provenance contract

Every registered run records identity, resolved configuration, command, environment, hardware, mathematical-object hashes, input hashes, metrics, certificates, logs, and artifact hashes. Historical runs remain immutable; deduplication creates a derived unique index rather than deleting records. The latest structural audit reports 150 historical runs, 9 unique scientific instances, 8 duplicate groups, 0 missing artifacts, and complete contracts for all 150 records. The duplicate warning is a governance issue, not a reason to reinterpret historical runs as independent evidence.

6. Paper-generation workflow

The v5 paper sources are under `papers/projected_graphs_v5`. They reuse registered v4 vector figures and bibliography while adding only claims supported by the V5 truth ledger. The build script compiles each source with `latexmk`, writes final PDFs to `output/pdf/`, runs `pdfinfo` and text extraction checks, and renders pages with Poppler for visual inspection. Generated auxiliary files remain in the intermediate build area and are not treated as scientific artifacts.

7. Governance and limitations

The current governance result is non-strict PASS/YELLOW. Required artifacts are present; warnings remain for duplicate historical runs and the paper quality rubric. The strict research gate remains fail-closed because theorem-level novelty, independent human review, repeated-law sharpness, global $k = 3$ sharpness, universal dimension/rank reduction, and global multilinear norm optimality are unresolved.

A compiled PDF is therefore a reproducible research artifact, not publication approval. No test-set or KGR claim is changed by this package.

8. Reproduction checklist

(1) Read `AGENTS.md`, the memory manifest, current state, tasks, and blockers.
(2) Inspect the relevant theorem and truth ledgers before changing a claim.
(3) Run the focused pytest command and JSON validation.
(4) Run governance audit and run deduplication.
(5) Compile the three papers and render every final PDF page.
(6) Record command, environment, branch, commit, changed files, and limitations in postflight.

9. Conclusion

The software package is experiment-ready and paper-build-ready, but not release-approved. Its primary value is that every mathematical statement, construction, test, artifact, and limitation has a durable location and a reproducible command path.

References

1. Nicholas J. Higham, *Accuracy and stability of numerical algorithms*, 2 ed., Society for Industrial and Applied Mathematics, 2002.
2. Ramon E. Moore, R. Baker Kearfott, and Michael J. Cloud, *Introduction to interval analysis*, Society for Industrial and Applied Mathematics, 2009.
3. Society for Industrial and Applied Mathematics, *Siam journal on scientific computing: Instructions for authors*, 2026, Accessed 29 July 2026.

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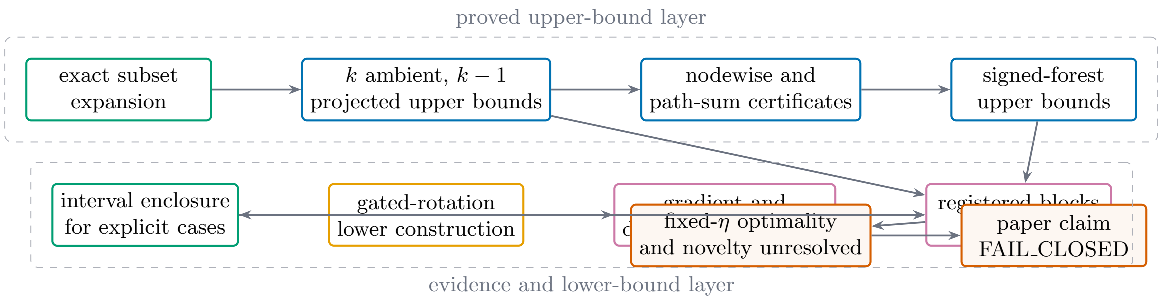

Figure 1. The evidence DAG separates proof, upper certificate, lower construction, empirical search, and release authorization.